

Lab Assignment 01

Intermediate Biomedical Data Science (EN.580.375)

Overview

This lab practices the concepts discussed Lectures 1–3. You will practice moving from measurements to vectors, probability models, likelihoods, optimization, and uncertainty.

Groups are allowed, but each student must submit their own code and short write-up. If you work with others, identify your collaborators and write your responses in your own words. Work on the lab across the MWF lab sessions after class (about one hour each day); the completed lab is due at the end of the week.

You may use Google Colab or a local Jupyter environment. The code chunks below are starter code: run and adapt them as needed, and include your code, plots, numerical results, and short explanations in the submitted notebook or equivalent file.

Suggested schedule

- **Monday:** Part 1 — vectors, projections, and data structure
- **Wednesday:** Part 2 — probability, distributions, sampling, and Bayes' rule
- **Friday:** Part 3 — likelihood, optimization, and uncertainty

Learning goals

By the end of the lab, you should be able to:

- represent observations as vectors and matrices
- use projections to measure alignment with a reference pattern
- calculate weighted averages and variances
- distinguish probability, likelihood, and posterior probability
- diagnose when a simple distribution is a poor model
- compare MLE and MAP estimates
- use gradient descent to minimize an objective
- use a bootstrap distribution to describe uncertainty

Part 1 — Vectors, projections, and data structure

1. A small biomedical dataset

The following matrix contains measurements from five observations on three features. You can imagine the columns as molecular or physiological measurements after preprocessing.

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array([
    [3.0, 2.0, 1.0],
    [4.0, 1.0, 2.0],
    [1.0, 3.0, 2.0],
    [2.0, 2.0, 4.0],
    [5.0, 0.0, 1.0],
])
```

Questions

1. What do the rows and columns of X represent? What does $X_{4,2}$ represent in words?
2. What is the shape of X ? What is the shape of Xv if $v \in \mathbb{R}^3$?
3. Explain why centering or scaling columns might be useful before comparing observations or finding directions of variation.

2. Projection onto a reference pattern

Let

$$\mathbf{x} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Treat $\mathbf{y}$ as a reference pattern involving the first two features.

Questions

1. Compute the projection

$$\mathbf{x}_2 = \text{proj}_{\mathbf{y}}(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{y}}{\mathbf{y}^T \mathbf{y}} \mathbf{y}$$

2. Compute the residual $\mathbf{x}_1 = \mathbf{x} - \mathbf{x}_2$.
3. Verify that $\mathbf{x}_1^T \mathbf{y} = 0$. What does this tell you geometrically?
4. What does $\mathbf{x}_2$ measure, and what information remains in $\mathbf{x}_1$?

3. Projection scores

Use the matrix X (defined in problem 1) and reference direction:

```
y = np.array([1.0, 1.0, 0.0])
```

Questions

1. Compute the projection coefficient for each row of X . Store the results in a one-dimensional array.
2. Which observation is most aligned with $\mathbf{y}$? Explain how you decided.
3. Plot the five projection coefficients with labeled axes and a descriptive title.

4. Explain why projection onto a direction can be useful for visualization or summarization.
5. The central dogma describes information flow from DNA to RNA to protein. Give one example of how a vector of measurements could represent one of these biological objects, and explain your thinking on if a projection score should be interpreted as an association with a pattern or as proof of a causal mechanism.
6. In a single-cell RNA-sequencing matrix, what do rows and columns usually represent? Give one reason cell-level normalization or gene filtering might be needed before computing projection scores.
7. Let $\mathbf{b}_1$ and $\mathbf{b}_2$ be two linearly independent reference directions. Explain the difference between their span and a basis for that span. How could projecting onto several directions relate to the SVD $X = U\Sigma V^T$?

4. Conceptual check: direction versus magnitude

Two observations have feature vectors $a = (1, 2, 1)$ and $b = (10, 20, 10)$.

1. What do you expect about their cosine similarity? Explain without calculating the exact value.
2. Would Euclidean distance and cosine similarity emphasize the same aspect of these observations? Why or why not?
3. Give one biomedical situation in which direction matters more than magnitude, and one in which magnitude is scientifically important.

Part 2 — Distributions, sampling, and Bayes' rule

5. A two-state weighted average and variance

Suppose a randomly selected cell comes from one of two states. State A occurs with probability 0.60 and state B occurs with probability 0.40. A marker is detected with probability 0.80 in state A and probability 0.20 in state B . Let $Y = 1$ when the marker is detected and $Y = 0$ otherwise.

Questions

1. Calculate the overall probability $P(Y = 1)$ as a probability-weighted average over the two states.
2. Calculate $E[Y]$ and $E[Y^2]$ from the two possible values of Y .
3. Use $\text{Var}(Y) = E[Y^2] - E[Y]^2$ to calculate the variance. Explain why the variance is not obtained by simply averaging the two state-specific probabilities.
4. How would the overall probability and variance change if state A became more common?

6. Poisson versus over-dispersed single-cell counts

Single-cell RNA-sequencing counts are often initially compared with a Poisson model, for which the mean and variance are both λ . Consider the following observed counts for one gene across cells:

```
gene_counts = np.array([0, 0, 1, 1, 2, 2, 3, 4, 8, 12, 15, 20])
```

Questions

1. Calculate the sample mean, sample variance, and Fano factor $s^2/\bar{x}$.
2. Does the Poisson mean-variance relationship appear plausible for these data? Explain what feature of the counts supports your answer.
3. Give at least two biological or technical reasons why single-cell counts might be over-dispersed relative to a Poisson model.

4. Why is over-dispersion a modeling concern for standard errors or likelihood-based conclusions?
5. Suggest one next step: for example, a negative-binomial model, a transformation, or a model with cell-level covariates. Explain what problem your choice is intended to address.

7. A Gaussian model for continuous measurements

Consider measurements of a normalized continuous biomarker:

```
biomarker = np.array([8.7, 9.4, 10.1, 10.3, 10.8, 11.0, 11.2, 11.9, 12.4, 13.1])
```

Assume $X \sim \mathcal{N}(\mu, \sigma^2)$ as a first approximation.

Questions

1. Compute the sample mean and sample standard deviation. What do these summarize?
2. Plot a histogram and overlay a Gaussian density using your estimated mean and standard deviation.
3. Use the empirical distribution or the Gaussian approximation to estimate the probability that a future measurement is below 10.
4. Identify one reason the Gaussian model may be useful here and one reason it may be inadequate.
5. Explain how the Central Limit Theorem differs from the claim that the individual measurements themselves are Gaussian.

8. Conditional and marginal probabilities

Suppose a condition is present in 2% of a population. A screening test has 90% sensitivity and 95% specificity.

Questions

1. Construct a two-by-two table for 10,000 people showing condition status and test result.
2. Calculate $P(\text{condition} \mid \text{positive})$.
3. Identify the conditional probabilities corresponding to sensitivity and specificity. Why are they not the same as the predictive value?
4. Explain how the base rate affects interpretation of a positive test.

Part 3 — Likelihood, optimization, and uncertainty without a Poisson model

9. Combining evidence for a Bernoulli probability

Suppose each patient either responds or does not respond to a treatment. The observed outcomes are:

```
responses = np.array([1, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0])
```

Assume $Y_i \sim \text{Bernoulli}(p)$ independently, where p is the probability of response.

Questions

1. Write the likelihood $L(p \mid y_1, \dots, y_n)$ as a product of Bernoulli probabilities.
2. Rewrite the likelihood using only the number of responses $k = \sum_i y_i$ and the sample size n .
3. Derive the log-likelihood, dropping terms that do not depend on p :

$$\ell(p) = k \log p + (n - k) \log(1 - p) + \text{constant}$$

4. Calculate the MLE $\hat{p}_{\text{MLE}}$. Explain why it equals the observed response proportion.
5. Plot the likelihood or log-likelihood over $0 < p < 1$. What does the plot show about the estimate?

10. MLE versus MAP with a Beta prior

Use the response data from Question 9 and the prior $p \sim \text{Beta}(a = 2, b = 2)$, where

$$\pi(p) = \frac{1}{B(a, b)} p^{a-1} (1-p)^{b-1}, \quad 0 < p < 1$$

Questions

1. Show that the posterior kernel is proportional to

$$p^{a+k-1} (1-p)^{b+n-k-1}$$

2. Identify the posterior distribution as $\text{Beta}(a + k, b + n - k)$.
3. Calculate the posterior mean and the MAP estimate using

$$E[p \mid y] = \frac{a + k}{a + b + n}, \quad \text{mode}(p \mid y) = \frac{a + k - 1}{a + b + n - 2}$$

4. Explain why the MAP estimate differs from the MLE. When would the prior have less influence?
5. Describe how a stronger prior, such as $a = b = 20$, would change the posterior.

11. Objective functions and gradient descent

Consider the objective $J(\theta) = (\theta - 3)^2$.

Questions

1. What value of θ minimizes $J(\theta)$? How can you identify it from the shape of the objective?
2. Compute $J(1)$, $J(2)$, and $J(4)$. Which candidate is preferred?
3. Derive $J'(\theta)$ and write the gradient-descent update.
4. Starting from $\theta^{(0)} = 0$, perform five updates with $\eta = 0.1$. Record $\theta^{(t)}$ and $J(\theta^{(t)})$.
5. Repeat with $\eta = 1.1$. What changes? Explain how the learning rate affects stability and convergence.

12. Bootstrap uncertainty for a fitted response proportion

Use the response vector from Question 9. The fitted Bernoulli probability is the sample proportion.

Questions

1. Write code for a bootstrap procedure that resamples patients with replacement, refits the response proportion, and repeats.
2. Implement at least 5,000 resamples and plot the bootstrap distribution of fitted proportions. Mark the observed sample proportion.
3. Calculate a central 95% bootstrap interval using the 2.5th and 97.5th percentiles.
4. Interpret the interval in terms of repeated sampling. What does it not tell you?

5. What assumptions are being made when patients are resampled independently? Give one situation in which that assumption would be questionable.
-

Final synthesis

13. From data to a scientific conclusion

Choose one result from this lab and write a short paragraph addressing:

1. What scientific or measurement question were you asking?
2. What data representation or probability model did you use?
3. What quantity did you estimate or compare?
4. What uncertainty or limitation should accompany your conclusion?
5. What additional data would make the conclusion more reliable?

14. Reflection

Answer briefly:

1. Which concept was most intuitive, and why?
2. Which concept remains unclear?
3. Describe one connection between linear algebra, probability, and statistical inference that became clearer through the lab.

Submission checklist

- a notebook or equivalent file containing your code;
- responses to all questions;
- labeled and interpretable figures;
- numerical results with appropriate interpretation;
- a small results table summarizing selected estimates and uncertainty;
- a final synthesis paragraph; and
- a brief note identifying collaborators, if you worked in a group.